

UST-YUDOMA, Russian Federation

WMO: 310540

Lat: 59.183N

Lon: 135.150E

Elev: 212

StdP: 98.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.4	-47.5	-50.8	0.0	-48.9	-49.2	0.0	-47.4	4.2	-30.0	3.4	-31.3	0.0	90	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	30.1	19.1	28.1	18.3	26.0	17.2	20.9	26.8	19.4	25.8	18.2	24.3	1.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.0	23.8	17.2	12.6	21.9	15.8	11.5	20.7	61.8	26.7	56.6	26.1	52.5	24.3	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.9	3.6	DB	-51.0	33.3	2.0	1.7	-52.5	34.5	-53.6	35.5	-54.7	36.4	-56.1	37.6	(4)
(5)			WB	-51.1	24.1	2.0	3.0	-52.6	26.2	-53.7	28.0	-54.8	29.7	-56.2	31.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.7	-37.0	-31.3	-17.7	-3.1	8.0	15.5	18.4	15.1	6.4	-6.1	-25.4	-36.8	(6)	
	DBStd	20.99	6.06	6.29	7.10	5.80	3.87	3.70	3.56	3.71	4.38	6.40	8.84	6.61	(7)	
	HDD10.0	7097	1458	1156	860	394	87	4	0	4	123	499	1061	1452	(8)	
	HDD18.3	9582	1716	1389	1118	644	321	98	44	115	357	758	1311	1710	(9)	
	CDD10.0	632	0	0	0	0	24	170	261	161	16	0	0	0	(10)	
	CDD18.3	74	0	0	0	0	0	14	47	13	0	0	0	0	(11)	
	CDH23.3	1086	0	0	0	0	12	287	589	196	2	0	0	0	(12)	
	CDH26.7	321	0	0	0	0	1	71	197	52	0	0	0	0	(13)	
(14) Wind	WSAvg	0.9	0.3	0.5	0.9	1.2	1.5	1.2	1.0	1.0	1.1	1.0	0.5	0.3	(14)	
(15) Precipitation	PrecAvg	353	14	10	11	12	27	38	54	52	51	34	28	15	(15)	
	PrecMax	502	29	18	61	54	58	101	143	260	162	78	63	44	(16)	
	PrecMin	216	6	2	2	0	4	5	12	2	6	9	8	6	(17)	
	PrecStd	77	6	4	12	11	14	23	26	51	33	19	16	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-16.3	-9.7	4.6	14.4	25.1	31.4	33.3	30.8	23.1	8.5	0.0	-18.8	(19)	
		MCWB	-16.7	-10.7	-0.1	7.0	13.2	19.1	20.6	19.2	15.4	4.3	-1.2	-19.0	(20)	
	2%	DB	-22.6	-14.8	0.9	10.3	21.7	28.9	31.0	28.0	19.5	5.8	-3.6	-21.4	(21)	
		MCWB	-22.8	-15.3	-2.5	4.1	11.7	17.6	19.8	19.2	13.1	3.5	-4.3	-21.7	(22)	
	5%	DB	-25.8	-18.5	-2.3	8.2	19.0	26.8	29.0	25.4	16.7	3.5	-8.9	-24.3	(23)	
		MCWB	-26.0	-18.8	-4.7	2.9	10.3	16.6	18.8	17.7	11.7	1.7	-9.5	-24.5	(24)	
	10%	DB	-28.4	-21.8	-5.2	6.1	16.7	24.7	26.8	23.0	14.4	1.8	-12.9	-27.2	(25)	
		MCWB	-28.5	-22.1	-7.0	1.7	9.0	15.4	18.3	16.6	10.4	0.4	-13.4	-27.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-16.7	-10.7	0.2	7.2	14.2	21.4	24.5	21.9	16.5	5.7	-1.2	-19.0	(27)	
		MCDB	-16.4	-9.9	4.6	12.8	22.6	28.0	28.5	27.2	20.8	7.0	-0.2	-18.8	(28)	
	2%	WB	-22.8	-15.3	-2.3	4.8	12.3	18.9	21.6	19.8	14.1	3.7	-4.4	-21.6	(29)	
		MCDB	-22.6	-14.9	0.7	9.6	19.9	26.6	27.6	26.1	18.2	5.7	-3.6	-21.3	(30)	
	5%	WB	-26.0	-18.8	-4.5	3.2	10.9	17.5	20.1	18.4	12.4	1.9	-9.4	-24.5	(31)	
		MCDB	-25.8	-18.4	-2.3	7.7	18.1	24.7	26.3	24.1	16.1	3.2	-8.9	-24.3	(32)	
	10%	WB	-28.6	-22.0	-7.1	1.9	9.4	16.3	18.9	17.0	10.9	0.5	-13.4	-27.3	(33)	
		MCDB	-28.4	-21.8	-5.2	5.9	15.7	22.9	25.4	21.9	13.6	1.7	-12.9	-27.2	(34)	
(35) Mean Daily Temperature Range	MDBR		7.6	12.4	17.1	14.2	12.5	13.8	12.9	12.4	10.6	7.7	8.6	6.3	(35)	
	5% DB	MCDBR	9.3	14.1	18.2	15.0	17.8	18.2	17.4	16.9	15.9	8.4	9.3	7.5	(36)	
		MCWBR	9.2	13.7	14.9	9.8	9.5	9.0	8.3	8.7	9.9	6.0	8.8	7.2	(37)	
	5% WB	MCDBR	9.3	14.0	17.2	14.0	16.0	16.1	14.2	15.2	13.8	7.7	9.4	7.2	(38)	
		MCWBR	9.2	13.6	14.3	9.6	9.1	8.9	8.4	9.1	9.4	6.0	8.8	7.1	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.233	0.266	0.313	0.397	0.448	0.431	0.497	0.434	0.348	0.307	0.244	0.209	(40)	
	taud		2.027	2.079	2.025	1.893	2.040	2.224	2.030	2.203	2.410	2.262	2.040	1.960	(41)	
	Ebn at Noon		564	740	804	786	777	803	730	755	782	687	530	447	(42)	
	Edh at Noon		49	81	117	162	151	128	152	119	82	68	48	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.62	1.71	3.45	5.08	5.21	5.87	5.37	4.18	2.62	1.30	0.77	0.36	(44)	
	RadStd		0.03	0.06	0.12	0.25	0.46	0.45	0.37	0.38	0.18	0.17	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page